

ELE00136M Immersive and Interactive Audio

Assessment Documentation

This document outlines the work you must complete as part of the Immersive and Interactive Audio module, where you must design and implement an interactive musical system and virtual acoustic environment. In this document, the details of this task are given, along with specific guidance and the marking criteria. Completing this task will include the use of several software programs (MaxMSP, SketchUp, ODEON, MATLAB), and knowledge of a number of topics (HCI principles, ISO-3382 analysis, MIDI CC, acoustic modelling) that will be covered in the module's lectures and practical sessions.

1. Design and implementation of an interactive musical system and virtual acoustic environment	1
1.1. The assignment task	1
1.2. The interactive musical system	2
1.3. The interactive virtual acoustic environment	2
1.4. The technical report	2
1.5. The user guide	3
1.6. The video	3
1.7. Marking	4
2. General guidance	5
2.1. Support	5
2.2. Appendices	5
2.3. Academic writing	5
2.4. References	5
2.5. LaTeX	5
2.6. MaxMSP	5
2.7. SketchUp and ODEON	6
2.8. Video	6
2.9. The user guide	6
2.9. Submission	6
3. Marking Rubric	7

1. Design and implementation of an interactive musical system and virtual acoustic environment

This section outlines the assignment task and its component parts, including guidance on the different elements, details of the deliverables, and the mark scheme.

1.1. The assignment task

Your task is to design and implement a system in MaxMSP that allows a user to interact with a virtual acoustic environment by performing music in that space using some kind of musical instrument or system.

The musical system you make could take the form of a synthesiser, a drum machine, a sequencer or similar - exactly what you create is up to you.

You will use SketchUp to design a 3D model of a room-like environment which you will import into ODEON. ODEON is an acoustic modelling program which you will use to generate impulse responses (IRs).

You will then create an interactive virtual acoustic environment in MaxMSP using these IRs. This will require you to further develop your musical system to allow for interaction with the virtual acoustic environment (e.g. spatialised drum machine, quadraphonic auralised synthesiser, virtual orchestra).

You will write a technical report which covers the design and implementation processes, and offers a critical evaluation of your submission. You will also write a short user guide for the system and make a video demonstrating the features of the system you have made and the sounds it makes.

This task aims to test your technical skills in audio programming, acoustic modelling, and IR analysis and allows you the opportunity to apply your creative skills in the design of an engaging and interesting interactive experience.

Your submission will consist of a single .zip file containing:

- A PDF of your technical report.
- A PDF of your user guide.
- A .txt file with a link to your video demonstration.
- Your finished MaxMSP work (with supporting files),
- Your modelled IRs.

1.2. The interactive musical system

The musical system you develop could make use of different synthesis techniques (e.g. additive, subtractive, FM), sampling, and sequencing or any combination thereof. It should make use of MIDI, including MIDI CC for the control of different parameters. The design should demonstrate use of good Human Computer Interaction (HCI) principles.

1.3. The interactive virtual acoustic environment

The space you model should have at least 6 boundary surfaces with one internal structure. Simple cuboidal rooms will not suffice. An example would be a “fan-shaped” hall with a stage structure at one end (i.e. a basic concert hall design). You are free to experiment with whatever design you like, but take care not to spend too much time designing the 3D model.

Set up the acoustic modelling in ODEON by:

- applying appropriate material properties for all surfaces;
- defining source and receiver types and positions;
- calibrating simulation and output parameters.

Your finished system should make use of at least 2 IRs (each with a different combination of source and receiver locations). You can decide to make use of IRs in B-format or Binaural stereo.

You should conduct IR analysis on one of the results generated from ODEON (this must be conducted using a mono version of the IR), making use of the MATLAB Acoustic Parameter Toolbox. This should include analysis of RT60, EDT, Clarity and Definition. You will present and discuss these results in your technical report.

1.4. The technical report

Your technical report (PDF, max 10 pages, excluding appendices) should include the following:

- Introduction and motivation.
- MaxMSP design and implementation.
- Discussion of HCI principles.
- A summary of your SketchUp and ODEON modelling work.
- ISO-3382 IR analysis.
- A critical evaluation of the finished system.
- Conclusion.

To support your technical report you must submit the mono IR you used for analysis. If you wish you may also submit other audio files, for example: audio examples from your finished product, anechoic recordings and auralisation using your modelled IRs.

1.5. The user guide

You should write a short user guide for your system (max 2 pages). It should be clear and easy to read, outlining the key features, how they work, and how to use them. You may wish to include screenshots, flow diagrams, and/or a troubleshooting section.

1.6. The video

You should create a video demonstrating the features of your finished product. This should be 3-4 minutes long. Your video does not need to be complex, and should be clear and understandable. You should upload your video as Unlisted to YouTube and provide the link in a text file as part of your submission.

1.7. Marking

Criteria	Weighting
MaxMSP design & implementation: • Quality of sounds produced. • Demonstration of good HCI principles. • Use of MIDI. • Use of spatial audio. • Commenting and clarity of structure. • Technical achievement. • Quality of supporting files.	40%
Technical report: • Clarity, formatting, and structure. • Design and implementation. • Discussion of HCI principles. • ISO-3382 analysis • Quality of critical evaluation. • Quality of references.	40%
User guide: • Clarity, formatting, and structure. • Suitable writing style. • Use of diagrams and figures.	10%
Video: • Effectiveness of product demonstration. • Video production and sound quality.	10%
Total:	100%

2. General guidance

This section includes general information and guidance for this assessment.

2.1. Support

You should make sure that you start work on this assignment as early as possible and make use of the support available. You will have lab sessions dedicated to assignment work where you can receive guidance and formative feedback on your work. We will also make use of some of the module's workshop sessions to discuss this assignment.

2.2. Appendices

You can add relevant extra material to your report in the form of appendices, which are not counted in the page limit. An example of this is the inclusion of images you want to include that are too large for the body of the report. Do not include code listings in your reports or their appendices: any code should be submitted as separate files.

2.3. Academic writing

The library provides excellent resources presenting [guidance on academic writing](#). You should make sure your writing is clear, concise, and correct.

2.4. References

Your report should clearly reference other work that has informed your work, or from which you have sourced information and material. A list of these references should be included following the concluding section of your report. The references should be in [IEEE format](#).

2.5. LaTeX

It is strongly advised that you use LaTeX to prepare your report. An [Overleaf](#) template will be made available on the module's VLE page.

2.6. MaxMSP

Your submitted MaxMSP files should be well structured (making good use of patches and sub-patches). We will cover the fundamental techniques required to complete these tasks in the module's practical sessions, and it is up to you to conduct further research and your own work in order to develop your system in your chosen direction. The lab materials will provide you with further references to explore. The PCs in PT/310 have MaxMSP installed on them.

2.7. SketchUp and ODEON

Completing this assignment will require you to make use of SketchUp and ODEON, as well as the SU2Odeon plugin for SketchUp. You should take care to avoid spending too long on 3D and acoustic modelling work using these programs. Keep things simple and make sure that you adhere to the requirements outlined above. ODEON is only available in PT/310 so make sure that you plan your time accordingly.

2.8. Video

Your video should be clear and well-presented with good audio quality. You are not required to make use of sophisticated video editing techniques or software. Make sure your video has a clear voiceover and that it is sufficiently informative as a standalone video.

2.9. The user guide

Your user guide should be fun and colourful, offering the user a helpful (but not boring) overview of how to use the system you have designed. It is *not* a technical report, so feel free to keep the writing style informal and don't worry about academic references (though clearly the usual standards of academic integrity apply). You are free to produce this user guide in any format you wish, so long as it does not extend beyond the size of 2 pages of A4 (e.g. it could be an A3 poster, or an A6 booklet).

2.9. Submission

Submission will be via the VLE.

3. Marking Rubric

This section includes the marking rubric for this task.

Criterion	Weighting	Distinction	Merit	Pass	Fail
MaxMSP design & implementation	40%	A professionally produced system that is fun to use and makes high quality, engaging, sounds. There is significant evidence of strong technical development in MaxMSP far beyond the materials provided in the lab sessions. The design demonstrates good understanding and application of HCI principles. MIDI is used highly effectively. The IRs are well integrated into the system and present a fun and engaging virtual acoustic environment. The patches and sub-patches are well laid out and clearly commented. The supporting files are complete and well-organised, including the mono IR used in the ISO-3382 analysis.	A nicely designed system that makes high quality sounds and is easy to use. There is evidence of technical development in MaxMSP that goes beyond the materials provided in the lab sessions. There is some evidence of HCI principles having been applied in the design of the system, and MIDI is used effectively. The IRs work well and create an effective virtual acoustic environment. The patches and sub-patches are clear with suitable comments. The supporting files are included and mostly well-organised, and the mono IR used in the ISO-3382 analysis has been submitted.	A basic system that shows little or no evidence of technical development in MaxMSP beyond the materials provided in the lab sessions. There is only the barest evidence of HCI principles having informed the design. MIDI has been used. IRs have been included and they can be used to apply a virtual acoustic to the sounds made. The structure of patches and sub-patches is fine and there are some comments. The supporting files are complete (or are mostly complete) but could have been organised better. The mono IR used in the ISO-3382 analysis is included.	A broken or incomplete system that does not fulfill the brief, not even making use of the materials provided in the lab sessions. This could be due to a clear lack of consideration of any HCI principles, no use of MIDI, or faulty IRs. The structure of the patches and sub-patches is unclear or broken, and there are no comments. The supporting files are incomplete or poorly organised. The mono IR used in the ISO-3382 analysis is missing.
Technical report	40%	A well-formatted, clearly-structured, and well-written report. Excellent use is made of figures and equations. References are complete and correctly formatted. The design and implementation of the finished system is explained in excellent detail, making it clear what the intended end product was and why, and showing an excellent understanding of synthesis methods, acoustic modelling, and how HCI principles informed the development process. The report presents an exemplary	A mostly well-formatted, clearly-structured, and well-written report, with few errors. Generally good use of figures and equations, with some omissions or minor errors. References are mostly complete and correctly formatted. The design and implementation of the finished systems is covered in mostly good detail, showing a high level of understanding of sound synthesis, the use of acoustic modelling techniques, and HCI principles. The report presents a complete	A report that covers all of the required content, but that is occasionally unclearly structured or poorly written. An attempt has been made to include some appropriate figures and equations, but there are some errors and missing detail. Some references have been included but these are limited in number and/or not formatted correctly. The design and implementation of the finished system is just about covered, though some sections may be underexplained or incomplete. There is some	An incomplete, poorly-formatted, and/or badly written report. Little to no use of appropriate figures and equations. References are missing or severely limited in number and incorrectly formatted. The design and implementation of the finished system is not described in clear detail, with missing or incomplete sections discussing the sound synthesis methods used, the HCI principles employed, and the acoustic modelling techniques. The ISO-3382 analysis of the

		ISO-3382 analysis of the modelled IR, including suitable figures and tables, and the room parameter data is explained clearly with clear connections made to the features of the modelled space. The critical evaluation is detailed and fair, making sensible, specific, and actionable suggestions for future improvements.	ISO-3382 analysis of the modelled IR, including some suitable figures and tables. The room parameter data is explained in some detail. The critical evaluation is correct and offers some sense of understanding where the system could be improved, with sensible suggestions for future improvement.	evidence of knowledge of sound synthesis methods, HCI principles, and acoustic modelling, but this is limited and not clearly related to the design of the system. An attempt has been made to present an ISO-3382 analysis of the modelled IR, with some limited use of figures or tables. The acoustic parameter data is presented and described but not explained fully. The critical evaluation is there and what it says is correct, but is very limited and makes only general and vague suggestions for future improvements.	modelled IR is missing or incomplete. The critical evaluation is missing or very short and facile.
User guide	10%	A very entertaining user guide that makes use of engaging figures, tables, and other graphical design features and clear instructions to communicate to the reader what the product does and how to use it.	A well-laid out and clearly written user guide that makes good use of figures and tables (and other design features) to communicate clearly what the system does and how to use it.	A perfunctory user guide that offers a fairly clear overview of the features of the system, but makes little use of appropriate figures, tables, or other design features to make this guide both fun and useful.	An unclear or incomplete user guide that does not give a clear indication of what the system can do and how to use it.
Video	10%	A fun, very well-produced video that clearly and concisely demonstrated the features of the system, explaining what is going on with a high quality voiceover.	A very clear, well-edited, video that makes effective use of voiceover and examples from the system to clearly show the audience what has been designed and implemented.	A simple clear video that shows the capabilities of the designed system. There is little in the way of context or explanation of what is being demonstrated, but there is a voiceover that communicates some information to the audience.	The video is unclear or of low quality, with distorted or glitchy (or no) audio, and no clear context about what is being demonstrated. The voiceover is absent or poorly recorded or very unclear.